

Vijay Shankar M B

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SUMMARY

AI Engineer focused on building production-ready Generative AI systems using Python, FastAPI, LangChain, LangGraph, RAG pipelines, vector databases, and scalable backend architectures. Experienced in developing multi-agent AI systems, AI chat platforms, semantic retrieval pipelines, streaming infrastructures, and full-stack AI products with modern deployment workflows.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

AI / GenAI: LangChain, LangGraph, RAG, Vector Embeddings, Semantic Search, Prompt Engineering, LLM APIs, AI Agents, Retrieval Pipelines

Backend: FastAPI, Node.js, Express.js, REST APIs, Streaming APIs, JWT Authentication

Databases: PostgreSQL, MongoDB, ChromaDB, SQLAlchemy, Prisma ORM

Frontend: React, Next.js, Tailwind CSS

Developer Tools: Git, Postman, Docker, Vercel, Render, GitHub

PROJECTS

Multi-Agent AI Research System

Python, FastAPI, LangGraph, LangChain, Gemini API, PostgreSQL, Docker, Next.js, Tailwind CSS

- Architected a production-ready multi-agent AI research platform using LangGraph for agent orchestration, state management, and autonomous research workflows.
- Engineered AI agent pipelines capable of dynamic tool calling, real-time web search augmentation, contextual summarization, and structured report generation.
- Implemented streaming conversational AI with asynchronous FastAPI SSE endpoints enabling real-time token streaming and interactive research sessions.
- Built persistent multi-session conversational memory using PostgreSQL with contextual state restoration and long-running workflow support.
- Containerized the complete application stack using Docker for reproducible deployment workflows and environment portability.

AI Research Assistant

Python, FastAPI, LangChain, Tavily, ChromaDB, PostgreSQL, Gemini API, Next.js, Tailwind CSS

- Engineered a production-style AI research assistant with streaming conversational AI and persistent multi-session memory.
- Built Retrieval-Augmented Generation (RAG) pipelines for PDF-based document intelligence using semantic chunking and vector embeddings.
- Integrated AI tool-calling workflows enabling dynamic web search augmentation through Tavily API.

- Designed scalable FastAPI backend services with asynchronous streaming APIs and PostgreSQL-based chat persistence.

ResumeIQ

Next.js, React 19, TypeScript, Express.js, Gemini API, MongoDB, NextAuth, Tailwind CSS

- Built an AI-powered resume intelligence platform capable of analyzing PDF resumes and generating ATS compatibility scores.
- Integrated Gemini LLM pipelines to generate contextual AI recommendations and optimization insights.
- Developed resume-to-job-description matching workflows with semantic similarity scoring and missing skill detection.
- Implemented secure authentication and protected user session management using Google OAuth and NextAuth.

EDUCATION

Diploma in Computer Application

JSS College, Mysore